

Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	qwen3:8b

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Editorial Decision

MINOR REVISION

Weighted Score: 3.5 / 5.0
Confidence: 83%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	5.0	Excellent	■■■■■
Manuscript Structure	2.5	Weak	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	3.5	Good	■■■■■
Results & Evidence	3.5	Good	■■■■■
Discussion & Conclusions	3.5	Good	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	3.5	Good	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	2.0	Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 5.0 / 5.0 | Confidence: 100%

Strengths:

- [TITLE] The title is specific, informative, and includes key terms for discoverability.
- [ABSTRACT] The abstract follows a structured format and clearly states the research problem and results.
- [KEYWORDS] The keywords are comprehensive, include domain-specific terms, and avoid duplication of title words.
- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.

StructureReviewer

Manuscript structure and logical coherence

Score: 2.5 / 5.0 | Confidence: 80%

Strengths:

- Clear section mapping
- Sections are logically ordered with introduction, methodology, experiments, discussion, and conclusions

Weaknesses:

- Missing literature review section
- Section flow is disrupted by repeated content in 'discussion' and 'section_vi'
- Research problem and hypothesis are not clearly stated
- Objectives are not explicitly stated or matched with conclusions

Major Comments:

- The manuscript lacks a literature review section, which is essential for contextualizing the research problem and existing work.
- The repeated content in 'discussion' and 'section_vi' indicates a lack of coherence and logical flow.
- The research problem and hypothesis are not clearly stated in the introduction, which undermines the clarity of the study's purpose.
- The objectives are not explicitly stated or addressed in the conclusions, which weakens the logical coherence of the manuscript.

Minor Comments:

- Section titles are not consistently formatted or descriptive.
- Some sections contain incomplete or cut-off content, which affects readability and completeness.

Recommendations:

- Include a literature review section to provide context and justify the research problem.
- Revise the 'discussion' and 'section_vi' sections to ensure they are distinct and add unique value.
- Clearly state the research problem and hypothesis in the introduction.
- Explicitly define the objectives and ensure they are addressed in the conclusions.
- Ensure all sections are complete and logically ordered.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 4.0 / 5.0 | Confidence: 90%

Strengths:

- Clear integration of physical constraints into the neural network framework
- Multi-objective optimization formulation
- Experimental validation on a five-story building prototype

Weaknesses:

- Lack of detailed description of the exact neural network architecture
- No mention of specific optimization algorithms used (e.g., genetic algorithm, particle swarm optimization)
- No mention of preprocessing steps for the accelerometer data

Major Comments:

- The manuscript should provide more details on the exact neural network architecture and the specific optimization algorithms used.

Minor Comments:

- Clarify the exact formulation of the multi-objective optimization problem.
- Include details on the preprocessing of the accelerometer data.

Recommendations:

- Provide a detailed description of the neural network architecture and the optimization algorithms used.
- Include a section on data preprocessing and the source of the accelerometer data.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Use of Kruskal-Wallis and post-hoc Wilcoxon/Holm corrections for non-parametric comparisons
- Bootstrap confidence intervals for robustness
- RMSE, MSE, and coefficient of variation for performance evaluation
- Pareto/hypervolume indicators for multi-objective optimization

Weaknesses:

- Lack of effect size reporting
- No explicit verification of test assumptions
- Reproducibility details not provided
- No mention of power analysis or justification for 30 runs

Major Comments:

- The manuscript should explicitly verify the assumptions of the Kruskal-Wallis and Wilcoxon post-hoc tests (e.g., independence, homoscedasticity).
- Effect sizes should be reported to complement p-values and provide a clearer understanding of practical significance.
- Reproducibility details, such as random seeds and resampling parameters, should be included for transparency.
- A power analysis or justification for the 30 independent runs should be provided to ensure sufficient statistical power.

Minor Comments:

- Consider explicitly stating whether the coefficient of variation is interpreted correctly in the context of the study.
- Clarify the interpretation of hypervolume and spread indicators in relation to the optimization objectives.
- Ensure that the RMSE and MAE are interpreted in the context of the problem and not just reported as metrics.

Recommendations:

- Include a power analysis or justification for the 30 independent runs.
- Report effect sizes (e.g., Cohen's d, eta-squared) to complement p-values.
- Provide details on the random seed and resampling parameters for reproducibility.

- Verify the assumptions of the Kruskal-Wallis, and post-hoc tests (e.g., independence, homoscedasticity).
- Clarify the interpretation of hypervolume and spread indicators in the context of multi-objective optimization.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- The manuscript includes a sufficient number of figures, tables, and equations to support the claims made in the text.
- Equations are sequentially numbered and appear to be properly introduced with context.
- Tables and figures are referenced in the text before they appear, which supports clarity and coherence.

Weaknesses:

- The extracted text does not expose table bodies, so it is not possible to verify if the data is precise and consistent across the paper.

Major Comments:

- The manuscript should provide the full content of tables and figures to ensure that the data is precise, consistent, and self-explanatory.

Minor Comments:

- Consider adding more detailed captions for figures to ensure they are fully self-explanatory.

Recommendations:

- Include the full content of tables and figures to ensure data consistency and clarity.

ResultsReviewer

Results validity and empirical support

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- MSE, RMSE, CV, and hypervolume metrics are reported, providing quantitative evaluation of performance.
- Pareto analysis and baseline comparisons with GA and PSO are included, supporting the novelty of the EANN framework.
- 30 independent runs are mentioned, suggesting robustness of the experimental design.

Weaknesses:

- The results section is incomplete and lacks detailed statistical analysis of the performance differences between methods.
- The text does not provide specific values for MSE, RMSE, CV, or hypervolume, making it difficult to assess the significance of the results.
- The section is cut off mid-sentence, which prevents a complete understanding of the results and their implications.

Major Comments:

- The results section is incomplete and lacks detailed quantitative analysis of the performance metrics (e.g., MSE, RMSE, CV, hypervolume).
- The text is cut off mid-sentence, which prevents a complete evaluation of the results and their implications.

Minor Comments:

- The results section could benefit from more detailed descriptions of the statistical validation and significance of the performance differences.

Recommendations:

- Provide complete and detailed values for all quantitative metrics (MSE, RMSE, CV, hypervolume, etc.) and their statistical significance.

- Ensure the results section is fully completed and coherent, with all claims tied to the figures and tables.
- Include a detailed statistical analysis of the performance differences between the EANN framework and the baseline methods (GA and PSO).

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Discussion provides a clear comparison of EANN with GA and PSO in terms of performance metrics and statistical significance
- Conclusions are directly tied to the experimental results and the research hypothesis

Weaknesses:

- Discussion lacks a detailed comparison with prior work in the literature
- Future work directions are not clearly specified or justified
- Limitations of the study are mentioned but not elaborated on

Major Comments:

- The discussion should include a more comprehensive comparison with prior work to contextualize the findings within the broader field.

Minor Comments:

- The limitations section could be expanded to provide more depth on the study's constraints.

Recommendations:

- Include a detailed comparison with existing literature to contextualize the findings.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical terminology
- Logical flow between sections

Weaknesses:

- Overuse of passive voice
- Vague quantifiers
- Undefined acronyms

Major Comments:

- Passive voice overuse reduces clarity and engagement. Consider rephrasing sentences to use active voice where possible.
- The term 'significant' is used without numerical context, weakening the claim.
- The acronym EANN is introduced without definition, which may confuse readers.

Minor Comments:

- Some sentences are overly complex and could be simplified for better readability.
- The introduction lacks a clear research question or hypothesis.

Recommendations:

- Define all acronyms on first use.
 - Replace vague quantifiers with specific numerical values where possible.
 - Revise passive voice constructions to active voice for improved clarity and engagement.
-

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Recent references from 2021-2023
- Consistent citation style (IEEE format)

Weaknesses:

- Lack of seminal foundational works
- Missing key areas in literature review

Major Comments:

- The references section lacks seminal foundational works that are critical for understanding the theoretical basis of lumped mass-spring-damper models and structural health monitoring. This omission weakens the manuscript's literature review and contextualization of the work.
- The absence of classic papers on structural parameter identification and observer-based state reconstruction limits the depth of the review and the manuscript's ability to situate its contributions within the broader field.

Minor Comments:

- The references section includes some recent and relevant works, which is a positive aspect.

Recommendations:

- Include seminal foundational papers on structural health monitoring and lumped mass-spring-damper models.
 - Add classic references on observer-based state reconstruction and parameter identification to strengthen the literature review.
-

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **2.0 / 5.0** | Confidence: 80%

Strengths:

- Limitations are explicitly stated in the discussion, though they are not sufficiently detailed or specific.

Weaknesses:

- The manuscript lacks comprehensive reporting on ethical considerations, data availability, conflict of interest, funding, and author contributions, which are critical for research transparency and integrity.

Major Comments:

- The manuscript fails to report essential ethical and transparency information, which is required by COPE and Q1 journal standards.

Minor Comments:

- The limitations section mentions limitations but does not provide specific or detailed information about them.

Recommendations:

- The authors are strongly advised to include ethical approval details, data availability statements, conflict of interest declarations, funding disclosures, and author contributions in the revised manuscript.
- The limitations section should be expanded to provide specific, honest, and detailed information about the study's limitations.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] The manuscript lacks a literature review section, which is essential for contextualizing the research problem and existing work.
- [2] The repeated content in 'discussion' and 'section_vi' indicates a lack of coherence and logical flow.
- [3] The research problem and hypothesis are not clearly stated in the introduction, which undermines the clarity of the study's purpose.
- [4] The objectives are not explicitly stated or addressed in the conclusions, which weakens the logical coherence of the manuscript.
- [5] The manuscript should provide more details on the exact neural network architecture and the specific optimization algorithms used.
- [6] The manuscript should explicitly verify the assumptions of the Kruskal-Wallis and Wilcoxon post-hoc tests (e.g., independence, homoscedasticity).
- [7] Effect sizes should be reported to complement p-values and provide a clearer understanding of practical significance.
- [8] Reproducibility details, such as random seeds and resampling parameters, should be included for transparency.
- [9] A power analysis or justification for the 30 independent runs should be provided to ensure sufficient statistical power.
- [10] The manuscript should provide the full content of tables and figures to ensure that the data is precise, consistent, and self-explanatory.
- [11] The results section is incomplete and lacks detailed quantitative analysis of the performance metrics (e.g., MSE, RMSE, CV, hypervolume).
- [12] The text is cut off mid-sentence, which prevents a complete evaluation of the results and their implications.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] Section titles are not consistently formatted or descriptive.
- [2] Some sections contain incomplete or cut-off content, which affects readability and completeness.
- [3] Clarify the exact formulation of the multi-objective optimization problem.
- [4] Include details on the preprocessing of the accelerometer data.
- [5] Consider explicitly stating whether the coefficient of variation is interpreted correctly in the context of the study.
- [6] Clarify the interpretation of hypervolume and spread indicators in relation to the optimization objectives.
- [7] Ensure that the RMSE and MAE are interpreted in the context of the problem and not just reported as metrics.
- [8] Consider adding more detailed captions for figures to ensure they are fully self-explanatory.
- [9] The results section could benefit from more detailed descriptions of the statistical validation and significance of the performance differences.
- [10] The limitations section could be expanded to provide more depth on the study's constraints.
- [11] Some sentences are overly complex and could be simplified for better readability.
- [12] The introduction lacks a clear research question or hypothesis.

Specific Improvement Recommendations

1. Include a literature review section to provide context and justify the research problem.
2. Revise the 'discussion' and 'section_vi' sections to ensure they are distinct and add unique value.
3. Clearly state the research problem and hypothesis in the introduction.

4. Explicitly define the objectives and ensure they are addressed in the conclusions.
5. Ensure all sections are complete and logically ordered.
6. Provide a detailed description of the neural network architecture and the optimization algorithms used.
7. Include a section on data preprocessing and the source of the accelerometer data.
8. Include a power analysis or justification for the 30 independent runs.
9. Report effect sizes (e.g., Cohen's d, eta-squared) to complement p-values.
10. Provide details on the random seed and resampling parameters for reproducibility.
11. Verify the assumptions of the Kruskal-Wallis, and post-hoc tests (e.g., independence, homoscedasticity).
12. Clarify the interpretation of hypervolume and spread indicators in the context of multi-objective optimization.

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